Extreme Value Theory

1 Extreme Value Theory: An Introduction

Much of classical statistics is Focused upon accurate estimation of the ·average· value of a series (i.e., the mean) or the ·average' relationship between two or more series(the OLS regression line), and the central limit theorem is all about the sampling distribution of the means of a set of draws from a series.

However, in many situations it is not the average value that is of interest but rather an extreme and rare event. For instance, in climate forecasting applications, we might be interested in estimating the highest a tide is likely to be this year, or what might be the maximum amount of rainfall expected in a one-day period this spring. Using standard models based on an assumption that the underlying data follow a normal distribution is likely to lead to very inaccurate forecasts for such extreme events because the normal distribution often fits poorly to the tails of the distribution of actual observed values of a series. If we are interested in estimating the likelihood of *extreme* events rather than *typical* ones, it makes sense to use an approach that is focused on modelling the tails such as extreme value theory (EVT).

EVT became more widely adopted in finance from the 1 990s onwards as a result of the increasingly prevalent realisation that asset return distributions systematically depart from normality due to their fat tails. Assuming a normal distribution can therefore lead to severe underestimates of the probability of large price movements in either direction, and so importantly can lead to a vast underprediction of the probability of severe losses.

As an illustration, Levine (2009) provides an example based on estimating the parameters of an extreme value distribution using a set of percentage returns on monthly medium maturity A-rated corporate bonds from January 1980 to August 2008. It turns out that the worst return in the sample was -10.84%. The extreme value distribution calculates a probability of a monthly return as low as this or lower (i.e., more negative) during a 30-year period of 1.4%. Yet the corresponding probability if instead we had assumed that the bond returns followed a normal distribution is approximately 8×10^{-7}, which is more than 16,000 times smaller than the probability according to the extreme value distribution. This is an event, albeit a very rare one, which actually happened during the sample period and yet if the series were normally distributed it would be unexpected even in ten millennia.

It is inevitably very hard to estimate tail probabilities accurately since by definition there will be very few extremes and thus very little data upon which to base estimates of the parameters of a tail distribution but EVT will usually get us much closer than the normal distribution.

It is also worth stating that the reverse of the above argument applies: while EVT can provide more a more accurate characterisation of the tail behaviour of a series than a normal distribution, the former should only be applied to the tails and it will not provide accurate estimates nearer the centre of the distribution.

Before we proceed to some notation and relevant formulae, we need to note an important point. Clearly, distributions have two tails (the upper and lower ones), so we need to be careful about which tail we are referring to in any given application. Unfortunately, most textbook treatments present their derivations and models as applied to the upper tail, although in financial risk management it is the lower tail (comprising all of the observations on the biggest losses) that is usually of interest. Since the model is applied separately to each tail and the distribution of actual data is unlikely to be perfectly symmetric, the parameter estimates applied to the upper tail are probably not going to be the same as those arising from an identical estimation on the lower tail. However, fortunately it is easy to flip the data from one tail to another by simply taking the negative of all the data points in that particular tail. Hence the extreme losses become extreme profits, and vice versa. So this is not a big issue but we need to be careful in each case to ensure that we are referring to the tail that we are interested in. For the purposes of the remainder of this section, we will use the upper-tail notation although it should be clear in each case to which tail we are referring from the context.

There are two approaches to parameter estimation that can be adopted under the broad umbrella heading of EVT: the *block maximum* framework and the *peak over threshold* framework. Each of these will now be discussed in turn.

2 The Block Maximum Approach

To implement this technique, suppose that we have a series, y, of total length T (i.e., we have T observations). We would separate the overall series into m blocks of data, each of length n (thus $m \times n = T$). Let the maximum observed value of the series in each block be denoted by $M_k, k = 1, \ldots, m$. So we have the blocks

$$M_1 = max(y_1, y_2, \ldots, y_n)$$

$$M_2 = max(y_{n+1}, y_{n+2}, \ldots, y_{2n}) \tag{1}$$

$$M_m = max(y_{(n(m-1))+1}, y_{(n(m-1))+2}, \ldots, y_{mn})$$

The results in Fisher and Tippett (1928) and Gnedenko (1943) show that, provided some additional assumptions hold, the distribution of a normalised (rescaled) version of the maxima in each block (i.e., the distribution of $(M_1, M_2, \ldots, M_m)$) converges asymptotically to a generalised extreme value (GEV) distribution as $m, n \to \infty$.

There are only three classes of extreme value distributions into which the normalised M_k must fall: the Weibull, Gumbel and Frechét. The properties of each of these are discussed in Box 14.1 (p. 596) and their pdfs are plotted in Figure 14.1 (assuming for the plot that $\mu = 0$ and $\sigma = 1$ in all three cases, while $\xi = 0$ for the Gumbel, $\xi = -0.2$ for the Weibull and $\xi = 0.2$ for the Frechét).

A tricky issue is, for a given total amount of data T, how to split the data into blocks as we could either have more smaller blocks or a lower number of longer blocks. If the blocks are too long (i.e., a small number of long blocks), the number of maxima will be very small and estimation inaccurate leading to parameter estimates with high variance (high standard errors). On the other hand, if the blocks are

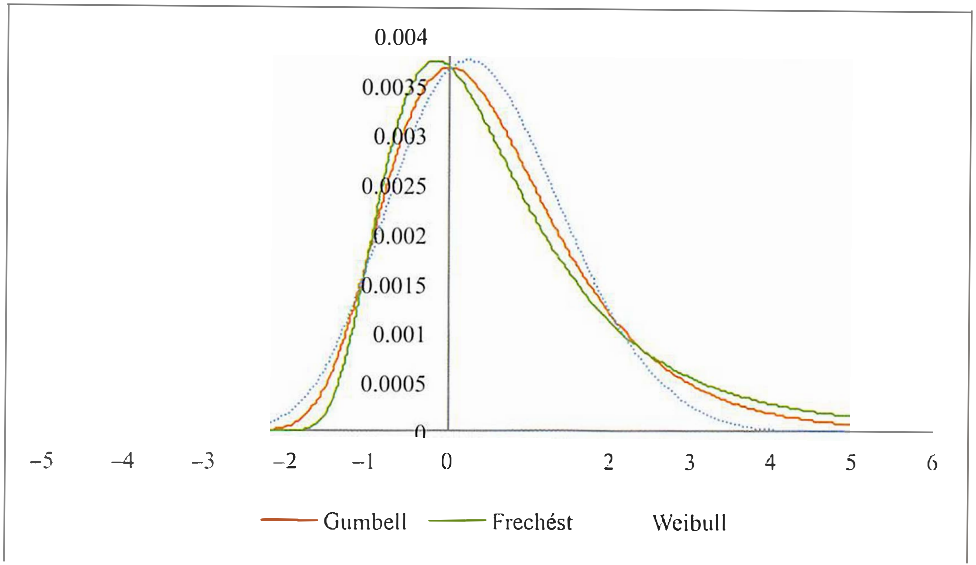

Figure 1 Pdfs for the Weibull, Gumbel and Frechét distributions

too short (a large number of short blocks), there is the potential for the maxima in some blocks to not be extreme values, which would cause a bias in the shape parameter estimate. Thus, there is a trade-off between bias and inefficiency here: long blocks = less bias but more inefficiency; short blocks = more bias but less inefficiency.

There are formal approaches that attempt to determine the block length optimally. But these are not discussed here since they are quite involved and in any case, it is clear that formally separating the data into blocks and using only the maximum value in each block is highly inefficient since there may be several extremes in some blocks and yet only the maximum will be counted in each case and the others ignored. An alternative method which avoids this arbitrary split of the data into blocks is known as the *peaks over threshold* (POT) technique. This approach has been preferred in most empirical applications and is considered in the following sub-section.

3 The Peaks Over Threshold Approach

Under this approach to extreme value modelling, an arbitrary high threshold U is specified, and any observed value of the series y_t exceeding this is defined as being in the extreme. Then in fact the exceedences over the threshold (call these $\tilde{y}_t = y_t - U|y_t > U$, where | means 'given') are actually modelled rather than the y_t themselves. As the threshold U tends to $(+/-)\infty$, the distribution of a normalised (rescaled)

BOX 14.1 The three generalised extreme value distributions

The cdfs for all three of the GEVs (Weibull, Gumbel and Frechét) can be described by the following equations (which are, more strictly, the limiting distributions as the sample size tends to infinity)

$$H_{\xi,\mu,\sigma}(y_t) = \begin{cases} exp[-(1+\xi(y_t-\mu)/\sigma)^{-1/\xi}] & \text{if} \quad \xi \neq 0 \\ exp[-exp(-y_t+\mu)/\sigma)] & \text{if} \quad \xi = 0 \end{cases} \qquad (2)$$

Note that H only exists for values of y_t that satisfy $1+\xi(y_t-\mu)/\sigma > 0$.

The GEV has three parameters that describe it: ξ is the shape parameter, which defines how fat the tails of the distribution are; μ is the location parameter; and σ is the scale parameter. Sometimes, the shape parameter is expressed in its inverted form, i.e., $1/\xi$, which is known as the *tail index*. Although it would perhaps be slightly misleading to call μ and σ the mean and variance of the distribution given that we are only focusing on the tail, they are analogous to the usual interpretations in describing the central tendency (where most of the data are located) and the dispersion (the spread), respectively. Which one of the three GEVs applies in any particular empirical application depends on the shape parameter, ξ

- The Frechét distribution applies when $\xi > 0$ and has fat tails. Thus it is the most appropriate class for modelling in finance to capture this empirical property of most financial time series that they are leptokurtic.
- The Gumbel distribution applies when $\xi = 0$ and has a medium thickness of tails which decay at an exponential rate.
- The Weibull distribution applies when $\xi < 0$ and has short tails with a finite end point – i.e., a fixed upper limit beyond which the pdf is exactly zero (and the cdf has already reached the value of one). This distribution would thus be appropriate for modelling the tails of some platykurtic series.

These three GEV distributions each encompass many other distributions as special cases. For example, the Frechét includes the Student t; the Gumbel includes the normal and log-normal; the Weibull includes some more specialist distributions such as the beta. In addition, we can view the Gumbel as the limiting distribution of the other two as the shape parameter tends to zero from above (for the Frechét) or from below (for the Weibull).

For some members of the GEV family, not all of the moments exist. To illustrate, in the context of the Student's t distribution with ν degrees of freedom, the kth moment only exists for $\nu > k$. So, for example, if $\nu = 3.5$ (corresponding to $\xi = 0.3$ approximately), only the first three moments would exist and none above that.

version of $\tilde{y}$ tends to the *generalised Pareto distribution* (GPD). We can write the cdf of the GPD as

$$G_{\xi,\sigma}(\tilde{y}_t) = \begin{cases} 1 - (1 + \xi \tilde{y}_t/\sigma)^{-1/\xi} & \text{if} \quad \xi \neq 0 \\ 1 - exp[-\tilde{y}_t/\sigma] & \text{if} \quad \xi = 0 \end{cases} \tag{3}$$

where again ξ is the shape parameter and σ is the scale parameter.*

As for the block maximum approach, there are three cases corresponding to the Weibull, Gumbel and Frechét distributions, respectively, which in the context of the GPD are the ordinary Pareto, exponential and beta distributions, respectively.

Furthermore, as above, the key parameter is ξ, which defines its shape. The first, when $\xi > 0$, corresponds to the fat-tailed case that is common for financial returns data. In fact, the tail index is the inverse of the number of degrees of freedom of the Student t distribution, ν, so that $\xi = 1/\nu$. Common estimates of ν are around 4–6, which would correspond to a tail index of 0.1–0.2.**

It should be clear that the generalised extreme value and the generalised Pareto distributions are closely related: in the limit in both cases, the former is the distribution of the standardised maxima while the latter is the distribution of the standardised data over a given threshold. The shape parameter, ξ, and therefore its inverse the tail index, are the same for both the extreme value and generalised Pareto distributions and thus the parameters from one approach can be obtained from an estimation under the other.

Choice of Threshold U

The choice of threshold U turns out to be important and, akin to the choice of block length, is tricky and involves a trade-off. If the threshold is too low in absolute value terms (and not far enough into the tail) then although this would provide a higher number of data points with which to estimate the distribution's parameters, some observations will be classified extreme when in fact they are not, leading to biased parameter estimates.

On the other hand, if the threshold is set too high, the number of data points classified as extreme will be small and thus although the bias in the estimates will be low, they will have a very high sampling variance as the effective number of observations being used to estimate the parameters will be small.

There are several approaches to reconciling these issues and selecting an 'optimal' threshold – some involve explicit calculations of the bias and variance, while others are more *ad hoc*. The simplest approach is to specify an arbitrary threshold that

*Note that β is often used to denote the latter parameter. In the context of $\tilde{y}_t$ being exceedences over a threshold, it would not make sense to also have a location parameter.

**We use the term 'tail index' when strictly they mean the shape parameter that we denote by ξ here.

places, for example, 1% of the distribution over the threshold. However, another fairly straightforward idea that is better matched to the data and seems to work reasonably well is to estimate the parameters using increasingly large values of U until the tail index estimate becomes stable (i.e., it stops changing as U is increased).

4 Parameter Estimation for Extreme Value Distributions

The parameters ξ, μ and σ can be estimated by maximum likelihood. As usual, this involves setting up a log-likelihood function based on an assumed distribution and then finding the parameter values that maximise it.

Equations (2) and (3) given above are both cdfs. To obtain the likelihood function, we would need to use the corresponding pdfs, take the natural logarithms and then sum them over the relevant sample observations. So first, to get the pdf, we would differentiate the function $G_{\xi,\sigma}(\tilde{y}_t)$ with respect to $\tilde{y}_t$. This would give the pdf for $\xi \neq 0$ as

$$g_{\xi,\sigma}(\tilde{y}_t) = \frac{1}{\sigma}\left(1 + \frac{\xi \tilde{y}_t}{\sigma}\right)^{-(\frac{1}{\xi}+1)} \quad (4)$$

Then the joint density for all N_U observations on $\tilde{y}_t$ over the threshold U will be the likelihood function given the data, which can be written as

$$LF(\xi, \sigma, \tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_{N_U}) = \prod_{i=1}^{N_U} g_{\xi,\sigma}(\tilde{y}_t) = \prod_{i=1}^{N_U} \frac{1}{\sigma}\left(1 + \frac{\xi \tilde{y}_t}{\sigma}\right)^{-(\frac{1}{\xi}+1)} \quad (5)$$

The log-likelihood function for a sample of N_U observations on the exceedences of a series y_t over a threshold U, $\tilde{y}_t$ (i.e., an upper tail estimation), is given by taking the natural log of the previous expression and expanding the parentheses

$$LLF(\xi, \sigma, \tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_{N_U}) = -N_U \ln(\sigma) - \left(\frac{1}{\xi} + 1\right) \sum_{i=1}^{N_U} \ln\left(1 + \frac{\xi \tilde{y}}{\sigma}\right) \quad (6)$$

where all notation is as above.

Provided that $\xi > -0.5$, maximum likelihood estimators are consistent and asymptotically normally distributed (Hosking and Wallis, 1987), and thus maximum likelihood estimators possesses these desirable properties. However, for extreme value distributions, unlike the normal case, there are no analytical solutions – in other words, there are no maximum likelihood formulae for the estimators given the data and thus a numerical approach (a search procedure) must be used instead. This is a significant disadvantage.

An alternative method is to directly estimate the tail parameter from the actual data using a *non-parametric* approach. A comparison between the two methods is provided in McNeil and Frey (2000). They show that, in some circumstances, the parametric approach may be preferable since it is applicable to a wider range of extreme value distributions and is less affected by the choice of threshold value. The non-parametric approach is much simpler to implement, however, since it does not require any optimisation.

The most common non-parametric approach is the Hill (1975) estimator, which is straightforward to implement and is applicable when the Frechét distribution is the relevant one (i.e., it only works for fat tails and not thin tails). Under some assumptions, the Hill estimator is consistent and asymptotically normally distributed (see Dowd, 2002; Rocco, 2011). The Hill estimator requires the raw data over the threshold $\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_T$ to be ordered. Note that the notation becomes much simpler if we reverse order the sample from largest first to smallest last (following, for example, Brooks, Clare, Dalle Molle and Persand, hereafter BCDP, 2005) rather than the standard smallest-to-largest ordering in most of the literature. So if we order the data on $\tilde{y}$, the exceedences over the threshold, from the largest, $\tilde{y}_{(1)}$, to the smallest, $\tilde{y}_{(T)}$, thus $\tilde{y}_{(1)} \geq \tilde{y}_{(2)} \geq \cdots \geq \tilde{y}_{(T)}$, then the Hill estimator of the shape parameter ξ is given by

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} [\ln(\tilde{y}_{(i)}) - \ln(\tilde{y}_{(k)})] \qquad (7)$$

where k is an integer to be selected, equal to the number of observations identified as being in the tail and is akin to the number of data points N_U exceeding the threshold of the POT method.

The most common approach to select k is again to estimate ξ over a range of plausible values of k and then to select the lowest value of it for which the estimate of ξ becomes stable. A graph of $\hat{\xi}$ against k is known as a *Hill plot.* Note also that in equations (7) and equations (8) and (9), if we wanted to calculate the tail index, ν, rather than the shape parameter ξ, we would need to take the inverse of the expressions $[.]^{-1}$ in each case.

Recalling the law of logs that $\ln(A) - \ln(B) = \ln(A/B)$, we can see that effectively the Hill estimator is taking an average over all observations within $k-1$ of the logs of the ratio of the value of one extreme point to the next most extreme one. In other words, it is estimating how quickly the tails fade away (the gradient of the pdf in the tail).

Two alternative non-parametric estimators that are effectively variants of the Hill estimator and that also estimate the speed of tail decay are due to Pickands (1975) and De Haan and Resnick (1980). The Pickands estimator is given by

$$\hat{\xi} = \frac{1}{\ln(2)} \ln\left(\frac{\tilde{y}_{(k)} - \tilde{y}_{(2k)}}{\tilde{y}_{(2k)} - \tilde{y}_{(4k)}}\right) \qquad (8)$$

The Pickands estimator is also consistent and asymptotically normal but is less efficient than the Hill estimator (Dowd, 2002, p. 212). The De Haan and Resnick estimator is given by

$$\hat{\xi} = \frac{\ln(\tilde{y}_{(1)}) - \ln(\tilde{y}_{(k)})}{\ln(k)} \qquad (9)$$

In order to provide a non-parametric tail estimator that removes small sample biases in the estimation of ξ, Huisman et al. (2001) develop a second stage regression of the estimates of ξ on a constant and the corresponding value of k used to estimate it

$$\hat{\xi}_i = \beta_0 + \beta_1 k_i + u_i \tag{10}$$

The modified estimate of ξ is then the intercept from this regression (β_0) which, from the basic definition of a regression intercept, is effectively the value that ξ will take in the limit as $k \rightarrow 0$.

Maximum likelihood estimation of the generalised Pareto distribution for a given set of data will yield estimates of both ξ and σ. However, it can be seen from the above that the non-parametric approaches estimate ξ directly, so what about σ (the scale parameter) – how should that be estimated? One approach would be to take the ξ estimate from the non-parametric (e.g., Hill) approach, plug this value into the log-likelihood function as a given constant and then use maximum likelihood to estimate σ as the sole remaining free parameter. This might result in the estimate of the scale parameter being more stable than if it were determined jointly with the shape parameter, but it hardly makes the use of the Hill estimator worthwhile if we then need to use maximum likelihood anyway.

BCDP use the result that if the $\tilde{y}$ follow a Frechét distribution, the scale parameter σ can be calculated given the shape parameter ($\hat{\xi}$, estimated from a non-parametric approach) and the data over the threshold as

$$\hat{\sigma} = \left(\frac{1}{k} \sum_{i=1}^{k} \tilde{y}_i^{1/\hat{\xi}} \right)^{\hat{\xi}} \tag{11}$$

5 Introduction to Value at Risk

Value at Risk (VaR) is a popular method for measuring financial risks – in that sense, it is a rival to others such a volatility (standard deviation), maximum drawdown, expected shortfall, etc. In general terms, VaR can be defined as an estimation of the financial losses which would be expected to arise from changes in market prices. More precisely, it is defined as the loss in monetary terms that is expected to occur over a pre-determined horizon and with a pre-determined degree of confidence. For example, a company might state that its one-day 99% VaR is ten million dollars. This would be interpreted as implying that the company is 99% confident that the maximum amount that it expects to lose on its portfolio of assets in a one-day period is ten million dollars.

VaR became highly popular as a risk measurement technique in the 1990s, and although its popularity has waned somewhat more recently and expected shortfall has emerged as the preferred approach, it is nonetheless worth studying as a good illustration of how extreme value distributions can be used in a practical finance setting. The evidence suggests that EVT can provide more accurate estimates of value at risk than the delta-normal method (both described below) since it provides a more precise characterisation of the shape of the tails of the distribution of losses, especially for extreme quantiles (such as the 99th percentile).

VaR became prevalent as a result of the simplicity of its calculation, its ease of interpretation, and from the fact that VaR can be suitably aggregated across an entire firm to produce a single number which broadly encompasses the risk of the positions of the firm as a whole.*

The calculated VaR is then used as a way to select the appropriate minimum capital risk requirement (MCRR), which is the value of liquid assets that a firm needs to hold in order to ensure that it can cover the expected losses should they materialise. It should be clear that the calculation of VaR and the corresponding selection of capital risk requirements involves a trade-off. If capital requirements are set too low, there is a danger that they will run out when needed most, resulting in financial distress or possibly even insolvency for the bank or securities firm involved. On the other hand, if the amount of capital is too high, then it is locked into an unprofitable usage as liquid assets including cash and Treasury bills usually provide very low returns.

There are several simple ways to compute VaR. The first is to assume that the distribution of portfolio losses is normal, and then we can simply take the appropriate critical value from the normal distribution at the α significance level multiplied by the standard deviation σ of the data $y_1, y_2, \ldots, y_N$

$$VaR_{normal} = \sigma Z_\alpha \tag{12}$$

To then obtain the VaR in cash value terms, we would multiply the figure resulting from this equation by the value of the portfolio. This is sometimes called the *delta-normal* method for VaR calculation. This approach gives equal weight to all observations in the sample period in calculating the standard deviation, σ. But it would also be possible to replace it by an estimate or forecast of σ from any other parametric model – such as an EWMA or GARCH model.

A second approach is to sort the portfolio returns and then simply select the appropriate quantile from the empirical distribution of ordered returns. This is a fully non-parametric approach that does not assume any specific distribution for the returns. This is arguably the simplest method for VaR calculation, and is sometimes termed 'historical simulation', a slight misnomer as a label that actually means to collect a sample of the historical returns (on the asset or portfolio under consideration), rank them and then take the fifth or first percentile of the empirical distribution. Then this number, multiplied by the value of the portfolio, is used as the VaR at the 95% or 99% confidence level, respectively. Historical simulation VaR is extremely simple to calculate, and often performs better than the delta-normal

* We need to note, however, that VaR is not *sub-additive*, meaning that the VaR of a portfolio is not a fixed combination of the VaRs of the component assets, and indeed in some circumstances the former can be larger than the sum of the latter.

approach since it can at least potentially capture the fat tails of actual distributions of losses. However, a key disadvantage is that the technique effectively uses only one data point and ignores all of the other information in the distribution of points that are both less and more extreme, whereas EVT uses all of the data points that are defined as being in the tail.

Using extreme value distributions to calculate VaR proceeds as follows, as summarised nicely in Jorion (2006, Chapter 10). Given the POT approach, assuming that we have estimated the parameters (ξ and σ) from the sample data, the cdf G in equation (3) would give the quantile α (from 0 to 1) corresponding to a particular value of $\tilde{y}$, $G(\tilde{y}) = \alpha$. We thus effectively invert the cdf to determine, given a quantile of the assumed extreme value distribution, what is the corresponding value of y and this would be the VaR

$$VaR = U + \frac{\hat{\sigma}}{\hat{\xi}}\left[\left(\frac{N}{N_U}\alpha\right)^{-\hat{\xi}} - 1\right] \tag{13}$$

Usually, $\alpha = 0.01$ or 0.05 would be of most interest, corresponding to 99% and 95% confidence, respectively. Note that the formula includes N/N_U, which is the ratio of the total number of observations in the whole sample (N) to the total number exceeding the threshold (N_U).

For the block maxima approach, VaR can be calculated in an analogous fashion as

$$VaR = \hat{\mu} + \frac{\sigma}{\hat{\xi}}\left[1 - (-m\ln(\alpha))^{-\hat{\xi}}\right] \tag{14}$$

for $\xi \neq 0$ and where m is the block length and all other notation is as above.

The VaR arising from the Hill estimator is calculated using the formula

$$VaR = \tilde{y}_{(k)}\left[\frac{N}{N_U}\alpha\right]^{-\xi} \tag{15}$$

where $\tilde{y}_k$ is the kth observation in the ordered series $\tilde{y}$, which is the same observation corresponding to the threshold when estimating ξ using formula (7) .

Again, we need slight care here since if we have adopted the convention of multiplying all the returns by -1 to turn the negative tail containing the losses into positives to simplify the calculations, then the appropriate values of α would be 0.99 and 0.95 and U would similarly be a positive number.

6 Some Final Further Issues in Implementing Extreme Value Theory

- It is possible to construct confidence intervals for VaR estimates arising from extreme value distributions, which would be useful to gauge whether they are precise or not. This is quite challenging to do in a valid and reliable way, however – see McNeil (1998).
- Embedded within the approach to estimating the parameters of an extreme value distribution is an assumption that the data are independently and identically

distributed. If the observations are not independent, this can lead to misleading estimates and therefore potentially inaccurate VaR calculations. Given the kind of dependence structure that is usually found in financial data, a common approach that seems to work is to estimate an ARMA-GARCH-type model, collect the standardised residuals (which are more likely to be iid than the original data) and then to estimate the parameters of the extreme value distribution on those – see Rocco (2011) and the references therein for further details.

- A very useful extension of EVT is to the multivariate case where, for example, joint distributions can be employed to measure common dependence and spillovers between extreme events in series. Here, copula functions are used to 'connect' the individual distributions for each series. Both the block maxima and POT approaches can be extended to the multivariate context. The multivariate extension, however, brings considerable increased complexity and there is not a unique definition of multivariate extremes. An additional problem is that, due to diversification or hedging effects, simultaneous extreme movements in several individual asset positions may not constitute an extreme for the portfolio comprising those assets as individual movements may be attenuated somewhat or even cancelled out entirely.

7 An Application of Extreme Value Theory to VaR Estimation

This section will now present an application of extreme value theory for the estimation of value at risk based on research in Gençay and Selçuk (2004). They employ several approaches (the delta-normal method, a Student's t distribution, historical simulation and EVT) to calculate and evaluate the VaRs for a set of emerging market equity index returns. They obtain daily data from Datastream for the following countries: Argentina, Brazil, Hong Kong, Indonesia, Korea, Mexico, Singapore, Taiwan and Turkey. The Philippines is also used for some of the analysis but not throughout. The sample period varies by country (presumably because of differences in data availability) but roughly covers the 1993–2000 period.

The returns for Brazil and Turkey have the highest standard deviations, but Hong Kong and Singapore have the highest kurtosis at 36.64 and 61.25, respectively. All series show excess kurtosis, suggesting that a fat-tailed distribution would be appropriate; almost all countries have considerable skewness as well, although strangely its sign varies. Their summary statistics also indicate that a daily loss of over 10% is perfectly possible, but this is up to 1000 times the daily standard deviation and thus a normal distribution would indicate that such a movement is so unlikely as to be almost impossible.

Gençay and Selçuk use the Hill estimator and other diagnostics to determine the appropriate threshold value that specifies whether a particular observation is classified as extreme or not. This information is presented here in Table 2, which also shows the corresponding quantile within the distribution for the threshold observation and the number of data points that lie in excess of this threshold

Table 2 Threshold percentage returns, corresponding empirical quantiles and the number of exceedences

	Threshold (%)	Quantile (%)	Exceedences (k)
Argentina	-2.7	6.7	129
Brazil	-3.8	7.0	130
Hong Kong	-7.0	0.6	41
Indonesia	-1.0	10.0	21
Korea	-3.5	4.5	130
Mexico	-3.0	5.0	69
Philippines	-4.0	1.8	19
Singapore	-2.5	2.6	101
Taiwan	-6.5	0.6	41
Turkey	-9.0	1.0	28

Source: Gençay and Selçuk (2004), Table 2. Reprinted with permission from Elsevier. Note that only the left tail results are reproduced here for brevity.

(the number of exceedences). In the interests of brevity and since it is of much more concern to most financial market participants (because they correspond to the extreme losses on long positions in those markets), we report only the lower tail part of their results here. The results show that the threshold is further out from the mean observation for Hong Kong (−7), Taiwan (−6.5) and Turkey (−9). In such cases, naturally the number of data points beyond that threshold is lower.

Table 3 presents the results from maximum likelihood estimation of the shape and scale parameters of the GDP and their associated standard errors. The former vary from a low of 0.03 for Korea and 0.15 for Brazil to a high of 0.60 for Taiwan. Inverting these numbers to obtain the tail indices gives figures of approximately 33.3, 6.67, and 1.67 respectively. This indicates that the Taiwanese series has particularly fat tails, with even the second moment (the variance) not existing ($\nu < 2$). The same is also almost true for Hong Kong and Singapore followed by the Philippines and Mexico. These results are slightly at variance with the summary statistics that Gençay and Selçuk present, where the Philippines has only a modest kurtosis and a small degree of skewness, perhaps attributable to the fact that the sample moments are estimated based on the whole distribution whereas the GPD parameters are based only on observations beyond the threshold.

The estimates of the scale parameter ($\hat{\sigma}$) vary from 0.5 and 0.6 for the Philippines and Indonesia to 1.8 for Brazil. Although again these estimates are based only on tail observations, in this case they do accord with the whole distribution estimate of the

Table 3 Maximum likelihood estimates of the parameters of the generalised Pareto distribution

	$\hat{\xi}$	SE($\hat{\xi}$)	$\hat{\sigma}$	SE($\hat{\sigma}$)
Argentina	0.20	0.01	1.1	0.2
Brazil	0.15	0.12	1.8	0.3
Hong Kong	0.48	0.22	1.6	0.4
Indonesia	0.32	0.09	0.6	0.1
Korea	0.03	0.11	1.5	0.2
Mexico	0.42	0.18	1.0	0.2
Philippines	0.44	0.37	0.5	0.2
Singapore	0.48	0.14	1.0	0.2
Taiwan	0.60	0.26	0.7	0.2
Turkey	0.22	0.27	1.6	0.5

Source: Gençay and Selçuk (2004), Table 3. Reprinted with permission from Elsevier. Note that only the left tail results are reproduced here for brevity.

standard deviation, which is high for Turkey and Brazil but low for the Philippines and Indonesia.

Gençay and Selçuk then perform an out-of-sample evaluation of the effectiveness of the VaRs chosen by the various approaches described above. They do this by determining the number of days for which the VaR estimate is violated (i.e., the actual return is more negative than the calculated VaR level). For a good model, we would expect that the percentage of out-of-sample violations (where the VaR is exceeded by the loss) should be roughly the same as the nominal confidence level embedded in the VaR calculation. So, for example, if we have a VaR estimated from a model with 99% confidence, we would expect the VaR to be sufficient on 99% of the out-of-sample test days.

Clearly, if the VaR is higher, we would expect fewer violations but there is a trade-off as described above: if the VaR is too high (an over-estimation of the risk) and thus capital requirements are too high, this conservative approach would lead to capital being tied up unnecessarily. On the other hand, if the VaR is too low (an underestimation of the risk), insufficient liquid capital would be held leading to the possibility of bankruptcy for the firm concerned. There may be an asymmetry within this trade-off, where the implications of insufficient capital are more serious than those of having too much.

Gençay and Selçuk use a rolling window for estimation with the VaR being estimated, for example, on observations 1 to 500, and then the level being compared to the actual return for day 501. The sample is then rolled forward one observation so

Table 4 Models that predict the actual left tail quantile most accurately

	5%	2.5%	1%	0.5%	0.1%
Argentina	H	N	E	E	E
Brazil	E	H	E	E	E
Hong Kong	H	T	E	E	E
Indonesia	N	T	E	E	E
Korea	T	T	T	T	T
Mexico	N	T	T	E	E
Singapore	E	N	E	E	E
Taiwan	N	T	T	T	E
Turkey	T	T	E	E	E

Source: Gençay and Selçuk (2004), Table 7. Reprinted with permission from Elsevier. Note that only the left tail results are reproduced here for brevity. H, N, T and E denote the situations where historical simulation, the delta-normal approach, the Student's t distribution and the GPD are the best model, respectively, for a given country and quantile, α.

that 2 to 501 are used for VaR estimation and 502 is used for testing and so on. They provide separate tables presenting the model, at each given confidence level and for each country stock index, which provides the least underestimation of the risk, and the model that provides the least overestimation of the risk. Finally, they provide a table that combines the two and presents the model with the VaR proportion of exceedences closest to the expected proportion given the confidence level. This latter table is shown here as Table 4.

The results in Table 4 clearly show that EVT becomes increasingly the approach of choice to calculate VaR as we move further and further into the tails. For VaR nominal exceedence level of 5% or 2.5% (corresponding to confidence levels of 95% and 97.5%, respectively), the Student's t distribution or historical simulation are the most successful overall, but at the 0.1% level (99.9% confidence), EVT performs best for all but one country. In fact, the separate underestimation and overestimation results (not shown here) demonstrate the superiority of EVT in both cases and thus we can conclude that it is more accurate than alternatives in capturing the shape of the tail of financial market return distributions, leading to better VaR estimation.

A further conclusion of Gençay and Selçuk is that the lower and upper tails have very different shapes and behaviour, and therefore it is necessary to use an approach that allows for this difference rather than one assuming a symmetric distribution. They find overall that models based on EVT outperform the more conventional approaches based on an assumption that returns follow a normal distribution or based on historical simulation (i.e., the actual quantile of the historical distribution of returns) for quantiles well within the tailsç for quantiles closer to the centre of the distribution - e.g., 50/o or 950/o, the winning approach is less clear-cut.